MILITARY SPECIFICATION

SPRAYER, POWERED, ULV, PORTABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers handcarried, electric-motor-driven/gasoline-engine-driven, one pint to one gallon ultra-low volume (ULV) pesticide sprayers.

1.2 Classification. The sprayers shall be of the following rated capacities (see 6.2):

Size 1 - one pint or 0.47 liter
Size 2 - one quart or 0.95 liter
Size 3 - two quart or 1.89 liters
Size 4 - one gallon or 3.79 liters

1.3 The sprayer shall be of the following type (see 6.2):

Type I - Gasoline-engine-driven
Type II - Electric-motor-driven

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Medical Bioengineering Research and Development Laboratory, ATTN: SGRD-UBE-V, Fort Detrick, MD 21701-5010, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	-	Treatment and Painting of Materiel.
MIL-S-12651	-	Sprayers, Packaging of.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-1472	-	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	-	Noise Limits for Army Materiel.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Reg. No. 241-208-AA-Cythion.

(Application for copies should be addressed to EPA, Registration Division, Product Control Branch, 401 M Street, SW, Washington, DC 20460.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The sprayer furnished under this specification shall be a product that has been tested and has passed the qualification inspection specified herein and has been listed on, or approved for listing on, the applicable Qualified Products List. One or more sprayers, as specified by the Government, shall be presented by the manufacturer for qualification inspection.

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3.2 Description. The sprayer shall consist of a housing, power unit, carrying handle, switch, pesticide dispenser, and pesticide reservoir. It shall be handcarried, operated by one person, capable of producing ULV spray droplets, weigh less than 28 pounds, permit variable flow of pesticide, and have a sufficient power source for continuous operation and uniform pesticide application for a minimum of 30 minutes.

3.3 Standard commercial product. The sprayer shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with these requirements. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.5 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.6 Design. Design of the equipment shall be in accordance with the best engineering practice and shall meet the criteria in MIL-STD-1472. The equipment design and accessory installation shall provide accessibility for maintenance and service in the field. Particular attention shall be given to features contributing to long periods of service without interruption for adjustment and repair. The design shall be such as to prevent conditions that may be hazardous to personnel or deleterious to equipment. Pesticide and gasoline filler caps shall be leak proof.

3.7 Power unit. The power unit shall be either an electric motor or a gasoline engine. Electric motors shall operate from rechargeable batteries and shall be capable of producing proper flow rates as specified for all pesticides registered by the EPA for ULV applications. The power unit must possess the reliability and maintainability as specified in 3.12 and 3.13.

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3.8 Pesticide reservoir. The pesticide reservoir and all components which may come in contact with pesticides shall be constructed of pesticide resistant materials. The reservoir shall be an integral part of the sprayer or may be a reusable pesticide cartridge or container. The reservoir shall be designed to form a leak-proof connection and shall be easily detached or designed for pesticide recharging without spillage. When filled to a level that will not involve pesticide spillage, the capacity of the reservoir shall not be less than the rated capacity for the particular size class.

3.9 Performance.

3.9.1 Spray range. The unit shall make effective deposition of spray solution (Reg. No. 241-208-AA-Cythion) at a spray range not less than 20 feet, when tested in accordance with 4.5.2.

3.9.2 Discharge rates. The unit shall be capable of a consistent flow rate which does not vary by more than 10 percent when tested in accordance with 4.5.3. The flow rate must be variably controlled with at least two settings within a minimum range of 0.1-0.3 ounces per minute and a maximum range of 1.0-1.5 ounces per minutes.

3.9.3 Particle size. The sprayer shall discharge droplets within a range of 1-50 microns, with 80 percent of the volume in the range of 5-30 microns when tested in accordance with 4.5.4.

3.9.4 Percent discharge. The sprayer shall discharge not less than 80 percent of the total rated reservoir capacity when tested in accordance with 4.5.5.

3.10 Weight. The net weight of the sprayer, and battery pack if applicable, shall be less than 28 pounds when tested as specified in 4.5.6.

3.11 Noise limits. The noise level of the sprayer shall not exceed 85 decibels (A-weighted) (dB(A)) when tested in accordance with 4.5.8 and MIL-STD-1474.

3.11.1 Precautions (noise). If the sound level exceeds 85 dB(A), chemical resistant labels shall be provided which state, "CAUTION - HEARING PROTECTION REQUIRED." Hearing protectors shall not be supplied with the sprayer.

3.12 Reliability. The sprayer shall operate for 200 hours with a minimum mean-time-between-failures of 50 hours when tested in accordance with 4.5.9.

3.13 Maintainability. The sprayer shall have a maintenance ratio of not more than 0.07 (see 6.5).

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3.14 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be properly lubricated before delivery and shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.15 Treatment and finish. Unless otherwise specified (see 6.2), the equipment shall be treated and finished in accordance with the manufacturer's best standard practice or in accordance with MIL-T-704.

3.16 Identification marking. The equipment shall be marked for identification in accordance with MIL-P-514 and MIL-STD-130.

3.17 Workmanship. A visual examination will be conducted to determine if the ULV portable sprayer is clean and free from sand, dirt, fins, pits, scale, rust, sharp edges, cracks, poor or unsound welds, or any other defect that could impair its performance or serviceability.

3.18 Safety. A chemical resistant black or yellow sign shall be permanently affixed in a prominent location in the sprayer stating "CAUTION - Comply with instructions and warnings listed on insecticide container prior to handling insecticide."

3.19 Optional items. When required (see 6.2), the following high mortality items shall be specified:

a. Nozzles.

b. Gaskets.

c. Bands and clamps.

d. O-rings.

e. Hoses.

f. Filters.

g. Spark plug (type I).

h. Contact points (type I).

i. Condenser (type I).

j. Electric motor (type II).

k. Batteries or power pack (type II).

l. Other parts which are identified as high mortality items.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

c. Packaging inspection (see. 4.6).

4.2.1 Qualification inspection. Qualification inspection shall be performed on one or more sprayers of each size specified. This inspection shall include the examination of 4.4 and the testing of 4.5. Failure of the sprayer to pass the examination or tests shall be cause for rejection. The sprayer may be a standard production unit from the contractor's current inventory provided the unit meets the requirements of the specification and is representative of the manufacturing technique applicable to the remaining units to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample selected in accordance with 4.3. This inspection shall include the examination of 4.4, the test and rejection criteria of 4.5, and testing of 4.5.1, 4.5.4, and 4.5.7.

4.3 Sampling. A random sample of sprayers shall be selected from each lot in accordance with MIL-STD-105.

4.3.1 Sampling for examination. Examination shall be based on inspection level II and an Acceptable Quality Level of 2.5 percent defective.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-3.

4.4 Examination. The sprayer and each sample selected in accordance with 4.3.1 shall be examined for compliance with the requirements specified in section 3 of this specification. This element of inspection shall encompass all visual examinations and dimensional measurements.

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4.5 Testing. The maintenance schedule of the operator's manual shall be followed during the test sequence. Failure to pass any phase of any test shall be cause for rejection. Leaks, loss of insecticide or chemical solutions, and any uncorrectable defect shall be causes for rejection.

4.5.1 Ruggedness. Drop the sprayer three times from a height of 3 feet onto a grass-covered ground area. The manner of dropping shall be such as to cause the sprayer to strike on entirely different positions in the three drops. Examine the sprayer to determine damage that would affect operation of controls and cause leaks; loosen handle and reservoir connections; break or misalign tubing, spray mechanism, and motor bearing supports; expose parts to accidental contact; and any other damage or distortion that could affect function in accordance with 3.9.

4.5.2 Spray range. Following the test of 4.5.1, charge the sprayer with spray solution (insecticide or mineral oil). Place four microscope slides, coated with polytetrafluoroethylene[1], or silicone coating (see 4.5.4), 20 feet from the spray mechanism or jet discharge nozzle. The sides shall be horizontal to the floor or ground. Operate the sprayer for not longer than 5 minutes with the discharge directed at the slides, and examine the slides for evidence of particle deposits. Absence of evidence of spray impact on all the slides shall be cause for rejection. If the sprayer cannot be repaired and pass a retest, the lot shall be rejected. This test shall be conducted when wind speed is less than 3 miles per hour (mph). The slides (one foot apart) shall be placed downwind from the jet discharge nozzle.

4.5.3 Discharge rates. Following the test of 4.5.2, add a measured volume of spray solution (insecticide or mineral oil) to the reservoir and operate for a fixed period of time. After completion of the dispersal period, measure the remaining volume of liquid and determine the discharge flow rate. Perform the test three times. Record individual values to determine conformance to 3.9.2. The test shall be conducted at low and high discharge rate settings.

4.5.4 Droplet size test. The sprayer shall be operated for 10 minutes, and the droplets collected using the following procedures of the Teflon(R) coated glass microscope slide-settlement chamber method as described below.

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[1] Teflon, registered trademark of E.I. duPont de Nemours & Co., Inc. The source of prepared slides will be available from the contracting officer.

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The sprayer shall comply with the requirements of 3.9.3. The spray solution shall be concentrated insecticide or mineral oil.

a. Use a 2 x 2 x 2 foot settlement chamber constructed of clean, unfinished plywood. The settlement chamber shall be airtight with a 2 x 2 foot sliding door on one side for aerosol entry. The chamber shall be elevated so that the door opening is centered with the nozzle(s) of the sprayer.

b. Place four Teflon(R) coated glass microscope slides on the floor of the settlement chamber for each aerosol collection. One slide shall be placed in the center of each of four quadrants of the floor area.

c. With the sprayer in operational mode, move in a straight line past the settlement chamber at a walking speed of 2 to 3 mph so that the nozzle(s) comes within 3 to 6 feet of the door opening at the closest point.

d. Ambient temperature shall be 70 to 100°F, and natural wind currents shall not exceed 4 mph while moving the sprayer past the settlement chamber.

e. Close door of the settlement chamber immediately after passage of the sprayer to prevent loss of the aerosol sample. Wait 15 minutes after closing door and then remove glass slides.

f. Flow rate for the test material shall be that which is produced by the maximum flow rate setting of the sprayer.

g. Repeat collection procedure one time.

h. Measure the diameter of 100 droplets per slide with minimum diameter measurement being the ocular division nearest 4 microns.

i. Correct measured diameters for spread on Teflon(R).

4.5.5 Percent discharge. Following the test of 4.5.4, fill the sprayer reservoir and spray until the liquid flow is inconsistent or ceases. Measure the amount of fluid remaining in the reservoir. Compare the amount of residual material to the total rated capacity of the sprayer. Determine conformance to 3.9.4.

4.5.6 Weight. Weigh the empty sprayer, and fully charged battery pack if applicable, to determine conformance to 3.10.

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4.5.7 Insecticide resistance of components. Following the test of 4.5.5, all parts that come in contact with insecticide during normal operations shall be immersed consecutively, for a period not less than 72 hours in each solution, in the following test solutions:

a. Three percent salt solution.

b. Ninety-one percent technical grade malathion.

c. Undiluted xylene.

d. Fuel oil (No. 2).

The parts and the used test solutions shall be examined to determine evidence of electrolytic action and chemical action that might impair safe operation. The sprayer shall then be reassembled and tested for spray range in accordance with 4.5.2. Any damaged part prior to reassembly shall be cause for rejection and replacement of the part tested. Ineffective performance after reassembly shall be cause for rejection.

4.5.8 Noise limits. Following the test of 4.5.7, measure the noise level at the ear of the operator when the machine is held in a normal operational position. The sprayer shall be operated to produce maximum noise conditions. Should the noise level exceed 85 dB(A) at the ear of the operator, failure to provide the precautions specified in 3.11.1 shall constitute failure to pass this test.

4.5.9 Reliability.

a. The electric-motor-driven (type II) sprayer shall be continuously operated for the first 50 hours without dispersing pesticides. During the 51-100 hour interval, the sprayer shall be operated to disperse spray solution (mineral oil) for not less than 15 minutes each hour of operation. During the 101-150 hour interval, the sprayer shall be operated without dispersing spray solution. During the 151-200 hour interval, the sprayer shall be operated to disperse spray solution for not less than 15 minutes each hour of operation. At the 100 and 200 hour points, the sprayer shall be tested for compliance with 3.9.1, 3.9.2, and 3.9.3. A failure for the purpose of this test is defined as any malfunction which causes inability to commence operation, cessation of operation, or degradation below the operating conditions required to continue the test, or causes a serious safety hazard to personnel or equipment.

b. The gasoline-engine-driven (type 1) sprayer shall meet the same criteria for reliability as the type II sprayer. Every 30 operating hours the engine and exhaust components of the sprayer shall be cleaned and decarbonized.

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4.5.10 Maintainability. Maintenance operations shall be performed during the 200 hour reliability test to determine conformance to 3.13.

4.6 Packaging inspection. The preservation, packaging, packing, and marking of the sprayer shall be inspected to the quality conformance requirements of MIL-S-12651.

5. PACKAGING

5.1 Preservation and packaging. The preservation and packaging shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A. The sprayers shall be preserved and packaged in accordance with level A of MIL-S-12651.

5.1.2 Level B. The sprayers shall be preserved and packaged in accordance with level B of MIL-S-12651.

5.1.3 Commercial. The sprayers shall be preserved and packaged in accordance with Commercial requirements of MIL-S-12651.

5.2 Packing. The packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. The sprayers shall be packed in accordance with level A of MIL-S-12651 as specified for boxed sprayers.

5.2.2 Level B. The sprayers shall be packed in accordance with level B of MIL-S-12651 as specified for boxed sprayers.

5.2.3 Commercial. The sprayers shall be packed in accordance with Commercial requirements of MIL-S-12651.

6. NOTES

6.1 Intended use. The hand-carried, electric-motor-driven/gasoline-engine-driven sprayers are intended primarily for indoor and outdoor control of insect vectors and pests by dispersing water-based emulsions and oil solutions of insecticides.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size of sprayer (see 1.2).

c. Type of sprayer (see 1.3).

d. When treatment and painting shall be other than as specified (see 3.15).

e. Levels of preservation, packaging, and packing required (See 5.1 and 5.2).

f. When a high mortality spare parts kit shall be required (see 3.19).

6.3 Contract data requirements. When this specification is used in an acquisition that incorporates a Standard Form 1423 (Contract Data Requirements List) and invokes the provisions of paragraph 52.227-7013 of the Department of Defense Federal Acquisition Regulation (DOD FAR) Supplement, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (SF 1423) incorporated into the contract. When the provisions of DOD FAR Supplement 52.227-7013 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 Qualification. With respect to products requiring certification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion on the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is Commander, US Army Medical Bioengineering Research and Development Laboratory, Fort Detrick, Frederick, MD 21701-5010, and information pertaining to qualification of products may be obtained from that activity.

6.5 Definitions. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total active operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded.

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6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools) and instructions on shipment of sprayers.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Review activities:

Army - MD, CE
Navy - MS, CG
DLA - CS

Preparing activity:

Army - ME

Project No. 3740-0117

Preparing activity agent:

US Army Medical Bioengineering
Research and Development Laboratory

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-53049	Sprayer, Powered, Ultra-low Volume (ULV), Portable

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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